

API & REST API Overview

1. What is an API?

An **Application Programming Interface (API)** is a set of rules and protocols that allows different software applications to communicate with each other. It defines how requests are made, how data is exchanged, and how responses are returned.

- Acts as a **bridge** between client (user/app) and server (data/service)
- Enables integration between systems (e.g., apps, databases, services)
- Commonly uses web technologies like HTTP

👉 Example: A weather app uses an API to fetch data from a remote server.

2. What is a REST API?

A **REST API (Representational State Transfer API)** is a type of web API that follows the principles of the Representational State Transfer architecture.

Key Characteristics:

- Uses **HTTP protocol**
- Stateless (each request is independent)
- Client-server separation
- Uses standard data formats (usually JSON or XML)
- Resource-based (everything is treated as a resource via URLs)

👉 Example endpoint:

<https://api.example.com/users/1>

3. HTTP Methods (Types of Requests)

REST APIs use standard HTTP methods to perform operations on resources:

GET

- **Purpose:** Retrieve data from the server
 - **Safe:** Yes (does not modify data)
 - **Example:**
`GET /users` → Fetch all users
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POST

- **Purpose:** Send data to create a new resource
 - **Safe:** No (modifies data)
 - **Example:**
`POST /users` → Create a new user
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PUT

- **Purpose:** Update an existing resource (replace entirely)
 - **Safe:** No
 - **Example:**
`PUT /users/1` → Update user with ID 1
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PATCH

- **Purpose:** Partially update a resource
 - **Safe:** No
 - **Example:**
`PATCH /users/1` → Update only specific fields
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DELETE

- **Purpose:** Remove a resource
- **Safe:** No
- **Example:**
`DELETE /users/1` → Delete user with ID 1

5. Why REST APIs are Popular

- Simple and scalable

- Platform-independent
 - Widely supported (web, mobile, cloud)
 - Easy to test and integrate
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Conclusion

APIs enable communication between systems, and REST APIs provide a standardized, efficient way to interact with web services using HTTP methods like GET, POST, PUT, PATCH, and DELETE. They are fundamental to modern web and mobile application development.